

AI Cooking Platform 2.0

Project Summary, Implementation Blueprint & PPT Material

Based on the uploaded AI Cooking Platform project brief, expanded with the requested public recipe community, Google Meet collaborative cooking, and temporary LLM double-check architecture.

1. Executive Summary

The AI Cooking Platform is an intelligent cooking ecosystem combining computer vision, recipe recommendation, generative AI assistance, social recipe sharing and collaborative cooking. The baseline project detects ingredients, recommends recipes, provides contextual cooking assistance and supports collaboration. The 2.0 scope adds a Reddit-like public recipe community and Google Meet-based cooking sessions.

2. Problem Statement

- Users often have ingredients but do not know what to cook.
- Traditional recipe platforms require manual input and provide limited context-aware assistance.
- Users need guidance while actually cooking, not only a static recipe.
- There is no integrated public layer for authentic user recipes, voting, discussion and creator following.
- Remote cooking with friends is usually separate from recipe discovery.

3. Objectives

- Detect ingredients from images using YOLOv8.
- Recommend and rank recipes using ingredient availability and user constraints.
- Use an LLM as a temporary second-opinion verifier for detection and recommendation.
- Keep the LLM removable so the core ML/recommendation system can eventually operate independently.
- Provide a context-aware cooking assistant.
- Build a public recipe community with posts, upvotes/downvotes, comments and creator following.
- Support collaborative cooking sessions using Google Meet.
- Use a modular architecture suitable for incremental implementation by AI/vibe-coding agents.

4. Final Feature Scope

Module	Scope	Priority
Ingredient Intelligence	Image → YOLOv8 → normalized ingredients	P0
Recipe Recommendation	Filtering + matching + ranking	P0
LLM Double-Check	Temporary validation layer	P0 / temporary
Cooking Assistant	Context-aware recipe help	P1
Public Community	Posts, votes, comments, follows	P0
Collaborative Cooking	Recipe sessions + Google Meet	P1
Moderation	Reports and basic moderation	P1

5. System Architecture

Layer	Responsibility	Implementation
Frontend	Detection, recipes, community, sessions	React; React Native later
Backend	Auth, business logic, orchestration	FastAPI
Computer Vision	Ingredient detection	YOLOv8 + OpenCV
Recommendation	Filtering, scoring, ranking	Python service
LLM Verification	Second opinion, never sole authority	LLM API behind provider interface
Assistant	Context-aware cooking help	LLM + structured recipe/session context
Database	Users, recipes, posts, social graph, sessions	MongoDB
Realtime	Presence and session events	WebSockets

Layer	Responsibility	Implementation
Meet integration	Create/schedule Meet-enabled sessions	Google Calendar API

Core flow

Image → YOLOv8 → ingredient normalization → recommendation engine → optional LLM verification → ranked recipes → selected recipe → context-aware assistant → community/session actions.

6. LLM Double-Check Architecture

The safest design is to keep the ML/rule pipeline authoritative and use the LLM as an advisory validator. The validator should receive structured outputs, return a validation state, and never silently overwrite the primary result.

- Store primary ML/rule result and LLM result separately.
- Return the primary result when the LLM is unavailable.
- Record disagreements for later model evaluation.
- Expose a configuration switch to disable the validator.
- Use a provider interface so the LLM can be removed without changing core business logic.
- Long-term target: make ingredient detection and recommendation independent of the LLM.

7. Public Community Design

Feature	Behavior
Recipe posts	Authenticated users publish authentic recipes with ingredients, quantities, steps, images and tags.
Voting	One active vote per user/post; switching vote changes the existing vote.
Comments	Comments and optional replies.
Following	Users follow/unfollow creators; followed creators can influence the feed.
Feed	Newest, top, hot and followed-creator views.
Profiles	Creator bio, recipes, follower/following counts.
Moderation	Report posts/comments and review reports.
Notifications	Comments, follows, vote milestones and session activity.

8. Data Model

Entity	Important fields
users	id, username, email, profile, created_at
recipes	id, title, ingredients, quantities, steps, tags, time, author_id
recipe_posts	id, recipe_id, author_id, body, images, score, created_at
votes	id, post_id, user_id, value (+1/-1), created_at
comments	id, post_id, author_id, parent_id, body, created_at
follows	id, follower_id, following_id, created_at
detection_runs	id, user_id, image_ref, ml_result, llm_result, final_result
recommendation_runs	id, user_id, ingredients, constraints, candidates, verification
cooking_sessions	id, host_id, recipe_id, status, start_time, meet_url, participants
notifications	id, user_id, type, actor_id, entity_id, read
reports	id, reporter_id, entity_type, entity_id, reason, status

9. API Blueprint

Area	Example endpoints
Auth	POST /auth/register, POST /auth/login, GET /users/me
Detection	POST /detect/ingredients, GET /detect/{run_id}
Recommendations	POST /recommendations, GET /recipes/{id}
Posts	POST /posts, GET /posts/feed, GET /posts/{id}
Votes	POST /posts/{id}/vote, DELETE /posts/{id}/vote
Comments	POST /posts/{id}/comments, GET /posts/{id}/comments
Following	POST /users/{id}/follow, DELETE /users/{id}/follow
Sessions	POST /sessions, POST /sessions/{id}/join, POST /sessions/{id}/start
Meet	POST /sessions/{id}/meet, GET /sessions/{id}/meet
Assistant	POST /assistant/message
Moderation	POST /reports, GET /moderation/reports

10. Implementation Plan for Vibe Coders / Antigravity

Do not ask an AI coding agent to build the whole application in one prompt. Give it a bounded phase, explicit contracts and acceptance criteria.

Phase 0 — Foundation

- Inspect repository and freeze scope.
- Create frontend/backend/ML/integration structure.
- Define environment variables, database schema and API contracts.
- Add health checks and seed data.

Acceptance: Application starts, backend responds, database connection works, contracts are documented.

Phase 1 — Auth + Recipe Domain

- Registration/login.
- Profile.
- Recipe CRUD.
- Recipe detail page.
- Authorization.

Acceptance: A user can securely create, edit, view and delete their own recipes.

Phase 2 — Ingredient Detection

- Image upload.
- YOLOv8 inference.
- Label normalization.
- Confidence and bounding boxes.
- Persist detection run.

Acceptance: A test image produces structured ingredient output and the UI displays it.

Phase 3 — Recommendation Engine

- Ingredient matching.
- Constraint filters.

- Ranking by match, missing ingredients and time.
- Transparent explanation.

Acceptance: API returns ranked recipes with reasons.

Phase 4 — LLM Double-Check

- Create VerificationProvider interface.
- Send structured ML/recommendation output to LLM.
- Store agreement/disagreement separately.
- Add timeout/fallback.
- Make feature configurable.

Acceptance: The system still works when LLM is disabled or unavailable.

Phase 5 — Cooking Assistant

- Recipe/session context.
- Current-step state.
- Chat endpoint.
- Conversation persistence.

Acceptance: Assistant answers using the selected recipe and current cooking state.

Phase 6 — Public Community

- Create posts.
- Feed sorting.
- Upvote/downvote.
- Comments/replies.
- Creator profiles.
- Follow/unfollow.
- Notifications.
- Reports/moderation.

Acceptance: Users can publish, vote, comment, follow and discover creators.

Phase 7 — Collaborative Cooking

- Create session from recipe.
- Invite/share session.
- Generate Meet-enabled Calendar event.
- Store meeting URL.
- Track participants with WebSockets.

Acceptance: Host can start a session and participants can join the Meet workflow.

Phase 8 — Testing + Deployment

- End-to-end tests.
- ML evaluation set.
- API/security tests.
- Error handling.
- Dockerization/deployment.

Acceptance: Critical user journeys pass and deployment is reproducible.

11. Master Prompt for Antigravity / AI Coding Agent

You are implementing the AI Cooking Platform 2.0 specification.

Work one phase at a time. First inspect the repository and propose the plan for the requested phase.

Do not build the entire system in one pass.

Keep frontend, backend, ML inference, recommendation, LLM verification, community and integrations modular.

Define API/database contracts before wiring UI.

YOLOv8 and deterministic recommendation logic are primary; LLM verification is optional and must never be a single point of failure.

If the LLM fails or is disabled, return the primary ML/rule result.

Keep secrets in environment variables.

For every phase add validation, error handling and basic tests.

After each phase report changed files, APIs, DB changes, tests, known limitations and next phase.

Start with Phase 0 only.

12. Definition of Done

- Frontend, backend and database behavior are synchronized.
- Core detection/recommendation works without the LLM.
- Votes are unique and server-validated.
- Comments and follows are authorization-checked.
- Meet links are stored and access-controlled.
- Secrets are not committed.
- Critical endpoints have automated tests.
- Setup and deployment instructions are reproducible.

13. Evaluation Strategy

Area	Metrics
Ingredient detection	Precision, recall, mAP, per-class performance
Recommendation	Top-k relevance, ingredient match, missing-ingredient count
LLM verification	Agreement rate, false corrections, latency, failure rate
Community	Vote integrity, duplicate prevention, authorization
Assistant	Context relevance, latency, error/hallucination cases
Collaboration	Session creation, Meet-link success, join flow
System	Latency, error rate, resource usage, deployment reproducibility

14. Risks & Mitigations

Risk	Mitigation
Weak detection	Improve dataset, augmentation, thresholds and correction workflow.
LLM wrong validation	Treat LLM as advisory and preserve primary result.
Community abuse	Rate limiting, reporting and moderation.
Vote manipulation	Unique user/post constraint and server-side validation.
Google API complexity	Isolate integration and test separately.
Over-scoping	Strict phased delivery and acceptance criteria.
AI-agent drift	Keep architecture/contracts/prompts in /docs.

15. Google Meet Integration

A practical implementation is to create a Google Calendar event and request Google Meet conference data for that event.

Google's current documentation describes explicit Meet conference creation through conferenceData with conferenceDataVersion=1, and conference generation is asynchronous. This integration should therefore live behind a backend service with OAuth and explicit failure handling.

16. PPT Material — Slide-by-Slide

1. Title

AI Cooking Platform 2.0

Intelligent Cooking + Community + Collaborative Cooking

2. Problem

Ingredient-to-meal decision is difficult.

Static recipe platforms lack automatic ingredient understanding.

Users need contextual guidance.

Recipe discovery and social cooking are fragmented.

3. Proposed Solution

YOLOv8 detects ingredients.

Recommendation engine ranks recipes.

LLM provides temporary verification.

AI assistant guides cooking.

Community enables publishing, voting, comments and following.

Google Meet enables collaborative cooking.

4. Key Features

Ingredient detection

Personalized recipe recommendation

Context-aware assistant

Public recipe community

Upvote/downvote + comments

Creator following

Collaborative cooking

5. System Architecture

React frontend

FastAPI backend

YOLOv8 + OpenCV

Recommendation engine

LLM verification layer

MongoDB

WebSockets

Google Calendar/Meet integration

6. Ingredient Detection

Image upload → preprocessing → YOLOv8 → normalized ingredients.

Confidence is retained.

LLM is a second opinion, not the primary detector.

7. Recipe Recommendation

Ingredient matching

Constraint filtering

Ranking

Missing-ingredient explanation

Optional LLM plausibility check

8. Community Platform

User-created authentic recipes

Feed and discovery

Upvote/downvote

Comments and replies

Creator profiles

Follow/unfollow

Notifications and reporting

9. Collaborative Cooking

Create a session from a recipe.

Invite participants.

Create a Google Meet-enabled event.

Store the meeting link.

Keep recipe/session context in the platform.

10. LLM Double-Check Strategy

YOLOv8/rules remain primary.

LLM validates structured output.

Agreement/disagreement is logged.

Failure falls back to the primary result.

Validator can later be removed.

11. Data + API

Users, recipes, posts, votes, comments, follows, detection runs, recommendation runs, sessions and notifications.

REST APIs plus WebSockets.

12. Implementation Plan

Foundation → auth/recipes → detection → recommendations → LLM verification → assistant → community → Meet → testing/deployment.

13. Evaluation

Detection: precision, recall, mAP.

Recommendation: top-k relevance and ingredient match.

LLM: agreement and false-correction rate.

Community: vote integrity.

Collaboration: session/link success.

System: latency and reliability.

14. Security & Moderation

Authentication and authorization

Rate limiting

Unique voting constraints

Input validation

Reports/moderation

Secret management

Audit logs

15. Future Scope

Fine-tuned ingredient model

Remove LLM verification when ML is reliable

Nutrition-aware recommendations

Behavior-based personalization

Creator reputation

Native mobile app

16. Conclusion

A full cooking ecosystem combining AI perception, recommendation, guided execution, community and remote collaboration.

17. PPT Narrative

- Opening: start with the simple problem — users have ingredients but do not know what to cook.
- Explain the technical pipeline from image to ingredients to ranked recipes.
- Show the LLM as a temporary validation layer rather than a hidden dependency.
- Introduce the community layer as the social evolution of the recipe platform.
- Show collaborative cooking as the bridge from online recipe discovery to real-time cooking.
- Close with the long-term goal: a modular platform that can reduce LLM dependency as specialized ML becomes stronger.

18. Source & Scope Note

The baseline project description is the uploaded AI Cooking Platform project brief. The public community, Google Meet workflow and temporary LLM double-check are proposed 2.0 extensions requested for this version.